PhoneQubit: a qubit in your hand

What each gesture does, the rules behind it, and six experiments

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The phone holds one qubit. Gestures apply rotations and measurements; the app draws each outcome with the quantum probability. The guide assumes you know kets, superposition and the Born rule. Open `https://praveen2600.github.io/phonequbit/` in Safari or Chrome, tap Start, allow motion, and turn on the phone's rotation lock.

1 The state and the axes

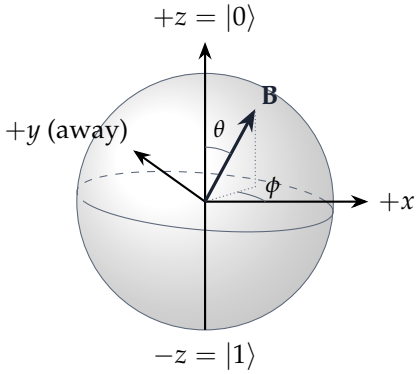
The state is

$$|\psi\rangle = \cos\tfrac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\tfrac{\theta}{2}\,|1\rangle,$$

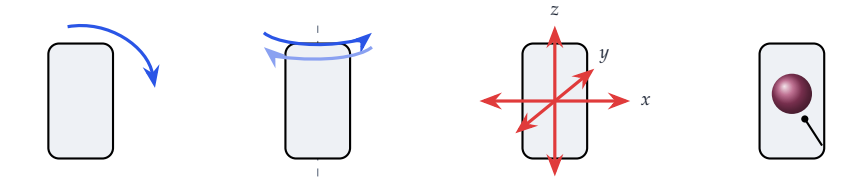
with Bloch vector $\mathbf{B} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$.
On screen. The ball is red for $|0\rangle$ and blue for $|1\rangle$, mixed in proportion to $P(1)$. Below it: θ, and the phase ϕ when the state is away from the poles. More → State shows $P(0)$, $P(1)$, ψ and a 3D Bloch sphere.
Axes are fixed in the room, seen with the screen facing you:

- z: up–down. $+z = |0\rangle$, $-z = |1\rangle$.
- x: left–right. $+x$ to your right.
- y: toward–away. $+y$ away from you.

These form a right-handed set. A jab measures along a whole axis, so the sign of y only fixes labels.



2 The controls



Twist steering wheel, in the screen plane

Flick quick door turn about a vertical line, and back

Jab quick straight push and stop

Tap tap the ball

Control	Operation	What it does
Twist	$R_y(\alpha) = e^{-i\alpha Y/2}$	Rotates **B** about y (the toward–away axis). Clockwise is positive: a quarter turn takes $\|0\rangle \to \|+x\rangle$. Changes θ; leaves $\|\pm y\rangle$ unchanged. Modes: 90° steps (default; a step happens at 55°) or smooth angle; tap the chip under θ to switch.
Flick	$R_z(\pm 90°)$, i.e. S or $S^\dagger$	Rotates **B** about z by ±90°: the phase ϕ jumps by ±90°, $P(0)$ and $P(1)$ stay the same. The two flick directions give opposite signs; the phase line shows which.
Jab	measures X, Y or Z	Projective measurement along the axis of the push: left–right → X, toward–away → Y, up–down → Z. Push direction (±) does not matter. Outcome ±; afterwards $\mathbf{B} = \pm\hat{\mathbf{n}}$. In 90°-steps mode the result comes 0.4 s after the stop (a twist in that time cancels the jab).
Tap	measures Z	Outcome 0 (**B** → $+z$) or 1 (**B** → $-z$). Same as an up–down jab.
Reset	prepares $\|0\rangle$	Also starts a new count: the measurement number and the Results line under the ball restart. The CSV export keeps every result, with a run number.
Repeat	prepares the last state again	Brings back the state from just before the last measurement, as a fresh copy from the same preparation. Needed because each measurement collapses the state. Off after Reset.
Pause / Go on	gestures off	Turning the phone no longer changes the state, so it can go back upright or lie on a table. Taps and Repeat still work; flicks and jabs need Go on.
Field pulse	$R_y(\omega t)$	More → Activities. Covering the front camera (or holding the button) switches on a field that turns **B** about y at $\omega = 45$, 90 or 180°/s. A pulse of length t turns it by ωt. Chips: Reset to $\|0\rangle$ (starts a new count), and Repeat.

Counting rule. A measurement counts when you hear the click and "Measurement N gave …" goes up.

3 The rules behind the outcomes

Measurement along a unit vector $\hat{\mathbf{n}}$:

$$P(\pm) = \tfrac{1}{2}\,(1 \pm \mathbf{B}\cdot\hat{\mathbf{n}}), \qquad \text{afterwards } \mathbf{B} = \pm\hat{\mathbf{n}}.$$

Special cases: $P(1) = \sin^2\frac{\theta}{2}$, independent of ϕ. For $\theta = 90°$: $P(+x) = \frac{1}{2}(1+\cos\phi)$ and $P(+y) = \frac{1}{2}(1+\sin\phi)$.

Gates from gestures (up to a global phase):

- quarter twist clockwise, $R_y(90°)$: $|0\rangle \to |+x\rangle$, $|1\rangle \to |-x\rangle$;
- two quarter twists, $R_y(180°) \simeq Y$: $|0\rangle \leftrightarrow |1\rangle$ (bit flip);
- one flick, S; two flicks, Z: $|+x\rangle \leftrightarrow |-x\rangle$;
- two flicks, then a quarter twist clockwise: $R_y(90°)\,Z = (X+Z)/\sqrt{2} = H$, the Hadamard gate. An anticlockwise twist gives a different gate, with $|0\rangle \to |-x\rangle$.

Statistics. With N repeats at probability p, the number of "1" results is $Np \pm \sqrt{Np(1-p)}$. For $N = 20$, $p = \frac{1}{2}$: 10 ± 2.2. About one run in 20 lands outside two standard deviations. A certain outcome ($p = 0$ or 1) must come out every time.

4 Six experiments

Saving effort. Use gestures only to prepare a state. Then tap **Pause**: the phone can go back upright or lie on the table, and taps and Repeat still work. For many runs, use **Repeat** then tap; the Results line counts them, starting from your last Reset. Flicks and jabs need **Go on**; hold the phone upright and still before a jab. Write your prediction before you measure.

1 Born rule

Do. Switch to smooth angle (tap the "90° steps" chip under θ). Reset, twist to any angle between about 30° and 150°, tap Pause, and note θ. Tap once; then Repeat and tap, 20 results in all.
Expect. $P(1) = \sin^2(\theta/2)$. For example 60°: 5 ± 1.9 ones in 20; 90°: 10 ± 2.2; 120°: 15 ± 1.9.
Physics. A single outcome is random; the frequency follows $|\langle 1|\psi\rangle|^2$.

2 Collapse and incompatible measurements

Do. Reset, quarter twist (any direction), Pause, bring the phone upright, Go on. Tap twice. Then do five rounds of: jab left–right once (measures X), then tap (Z).
Expect. The second tap repeats the first. The taps that follow a jab are random, about half 1.
Physics. A measurement leaves the qubit in the measured eigenstate. X and Z do not commute: learning X erases what you knew about Z. This is the three-magnet Stern–Gerlach chain z, x, z.

3 Can you see the phase?

Do. Prepare each state from Reset: quarter twist clockwise, Pause, bring the phone upright, Go on. Stop here for A; flick once for B (phase 90°), twice for C (phase 180°). For each state: tap once, then Repeat and tap 9 more times; then 5 jabs toward–away and 5 jabs left–right, with Repeat before each jab.

State	10 taps (Z)	5 jabs left–right (X)	5 jabs toward–away (Y)
A: $\|+x\rangle = (\|0\rangle + \|1\rangle)/\sqrt{2}$	50:50	always $+x$	mixed
B: $\|+y\rangle = (\|0\rangle + i\|1\rangle)/\sqrt{2}$	50:50	mixed	always $+y$
C: $\|-x\rangle = (\|0\rangle - \|1\rangle)/\sqrt{2}$	50:50	always $-x$	mixed

If B reads phase $-90°$, you flicked the other way and made $|-y\rangle$: the Y column then gives $-y$. Five equal results can happen by chance (1 time in 16); a certain result never shows the opposite sign.
Physics. $P(1)$ has no ϕ in it, so Z is blind to phase. X and Y ask how $|0\rangle$ and $|1\rangle$ add, and the answer depends on ϕ. **Bonus:** twist state B. Nothing changes, because $|+y\rangle$ lies on the twist axis.

4 Interference

Do. (a) Reset; quarter twist clockwise; Pause, bring the phone upright, Go on; quarter twist clockwise again; Pause. Tap once, then Repeat and tap 9 times. (b) The same, with one tap right after the first twist. (c) The same as (a), with two flicks after Go on, before the second twist.
Expect. (a) always 1. (b) random, about half 1. (c) always 0.
Physics. $R_y(90°)R_y(90°)|0\rangle = |1\rangle$: the two paths to $|0\rangle$ cancel. The tap in (b) destroys the superposition, and the cancellation is lost. Two flicks flip the relative sign, so the paths now cancel toward $|1\rangle$. Repeat prepares the final state again, so ten taps show whether it is certain or a superposition. (A left–right jab in the middle would change nothing: $|+x\rangle$ is an X eigenstate.)

5 Rabi oscillation with a field pulse

Do. More → Activities → Field pulse, rate 90°/s. Lay the phone in a lit place. Give one pulse of about 1 s (cover the camera), tap once, then Repeat and tap 9 times. Tap "Reset to $|0\rangle$" and do the same with pulses of about 2 s and 3 s. The app reports each pulse's length and angle.
Expect. $P(1) = \sin^2(\omega t/2)$, i.e. $\sin^2$ of half the reported angle: about 0.5 at 1 s, 1 at 2 s, 0.5 at 3 s.
Physics. A field rotates the spin steadily, so the pulse length sets the gate. A 1 s pulse here is a $\pi/2$ pulse ($|0\rangle \to |+x\rangle$), a 2 s pulse a π pulse ($|0\rangle \to |1\rangle$). Real qubits are controlled this way.

6 A hidden state (the seed of BB84)

Do. A partner prepares one of $|+x\rangle$ (twist clockwise), $|-x\rangle$ (twist anticlockwise), $|+y\rangle$ or $|-y\rangle$ (twist clockwise, Pause, upright, Go on, one flick either way). They tap Pause, turn on More → Settings → Hide the state, and hand you the phone. Hold it upright, tap Go on, and make *one* jab to find which state it is.
Expect. Jab along the right axis and you know for certain. Jab along the wrong axis and you get a random answer, and the state is now changed, so a second jab cannot help.
Physics. No single measurement distinguishes non-orthogonal states. An eavesdropper in the BB84 key-sharing protocol faces exactly this problem, and her mistakes give her away.

5 Stern–Gerlach mode, in brief

Switch to the Stern–Gerlach tab. You look along the atom beam; the phone is magnet 2, and its tilt φ (read from gravity) sets the field direction. Magnet 1 prepares the spin direction s ($\pm z$ or $\pm x$), or is off, which gives random spins from the oven. Each atom lands up or down with

$$P(\text{up}) = \cos^2 \tfrac{\varphi - s}{2},$$

the same rule as $P(0) = \cos^2(\theta/2)$ in qubit mode. Random spins give $\frac{1}{2}$ at every angle. A uniform field gives no splitting. "+ Magnet 3" builds the z, x, z chain of experiment 2. More → Lesson runs a guided five-step lesson.

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